AI-BASED EMPLOYEE ATTENDANCE MANAGEMENT SYSTEM
USING FACE RECOGNITION

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Introduction

The AI-Based Employee Attendance Management System Using Face Recognition is an intelligent attendance management application developed using Artificial Intelligence and Computer Vision technologies. Traditional attendance systems require manual entry or biometric devices, which are time-consuming and prone to errors. This project automates attendance by recognizing employees through facial recognition technology.

The system captures the employee's face using a webcam, compares it with stored facial encodings, identifies the employee, and automatically records attendance in the database. The application also includes employee management, attendance history, salary management, bonus and deduction management, and PDF report generation.

The project is developed using Python, Flask, OpenCV, Face Recognition Library, HTML, CSS, JavaScript, and MySQL.

KEY FEATURES

- AI-based Face Recognition
- Automatic Attendance Marking
- Secure Admin Login
- Employee Registration and Management
- Face Registration using Webcam
- Real-time Camera Attendance
- Attendance History Tracking
- Salary Calculation
- Bonus and Deduction Management
- PDF Report Generation
- User-friendly Dashboard
- Date and Time Display

Objectives

- To automate employee attendance using facial recognition.
- To eliminate manual attendance.
- To improve attendance accuracy.
- To prevent proxy attendance.
- To manage employee records efficiently.
- To maintain attendance history.
- To calculate employee salaries.
- To generate attendance reports in PDF format.
- To provide a secure login system.

Modules

Login Module

Provides secure administrator authentication.

Dashboard Module

Displays project statistics, employee count, attendance details, date, and time.

Employee Management Module

Allows the administrator to add, edit, view, and delete employee records.

Face Registration Module

Captures employee face images and stores them for future recognition.

Camera Attendance Module

Recognizes employee faces and automatically marks attendance.

Attendance History Module

Displays attendance records including check-in and check-out times.

Salary Management Module

Calculates employee salary using attendance, bonus, and deductions.

Reports Module

Generates attendance reports in PDF format.

System Architecture

Administrator
↓
Login Module
↓
Dashboard
↓
Employee Registration
↓
Face Registration
↓
Store Face Dataset
↓
Camera Attendance
↓
Face Recognition
↓
Attendance Database
↓
Salary Management
↓
PDF Report

Explanation

The administrator logs into the system and accesses the dashboard. Employee details and face images are registered. During attendance, the webcam captures the employee's face and compares it with stored facial encodings. If a match is found, attendance is recorded automatically. Salary is calculated using attendance records, and reports are generated in PDF format.

IMPLEMENTATION

Technologies Used

Frontend
HTML5
CSS3
JavaScript
Jinja2 Templates

Backend
Python
Flask

Database
MySQL

AI Libraries
OpenCV

Face Recognition
NumPy

Report Generation
ReportLab

Development Tools
Visual Studio Code
Git
GitHub

IMPLEMENTATION (CODING)

Face Registration Code

```
"image = face_recognition.load_image_file(image_path)
encodings = face_recognition.face_encodings(image)

if len(encodings) > 0:
    known_encodings.append(encodings[0])"
```

This code generates facial encodings from employee images and stores them for recognition.

Attendance Marking

```
"cursor.execute("""
INSERT INTO attendance
(emp_id, attendance_date, check_in, status)
VALUES (%s, CURDATE(), CURTIME(), 'present')
""", (emp_id,))"
```

This code automatically records employee attendance when the face is recognized.

Salary Calculation

```
"employee["net_salary"] = (
employee["salary_payable"] +
employee["bonus"] -
employee["deduction"]
)"
```

This code calculates the employee's final salary.

Dashboard Route

```
"@app.route('/dashboard')
def dashboard():
    if 'user' not in session:
        return redirect(url_for('login'))
    return render_template("dashboard.html")"
```

This route loads the dashboard page after successful login.

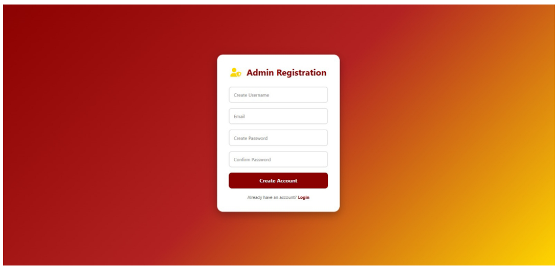
Login authentication

```
"@app.route('/login', methods=['GET', 'POST'])
def login():

    if request.method == "POST":

        username = request.form["username"]
        password = request.form["password"]

        conn = get_db_connection()
        cursor = conn.cursor(dictionary=True)

        cursor.execute(
            "SELECT * FROM users WHERE username=%s AND password=%s",
            (username, password)
        )

        user = cursor.fetchone()

        if user:
            session["user"] = username
            return redirect(url_for("dashboard"))

        flash("Invalid Username or Password")

    return render_template("login.html")"
```

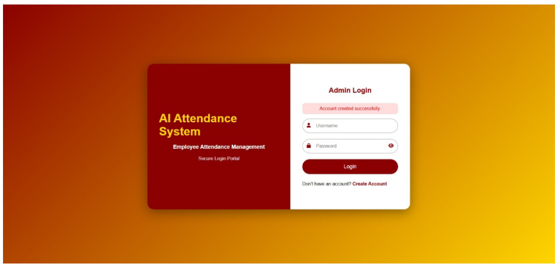
Verifies administrator credentials and redirects to the dashboard after successful login.

OUTPUT SCREENSHOTS

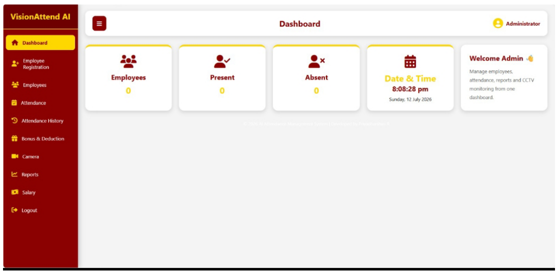
Admin Registration



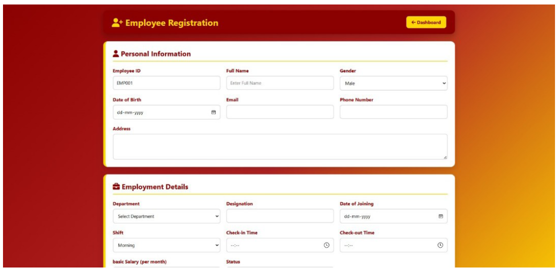
Login Page



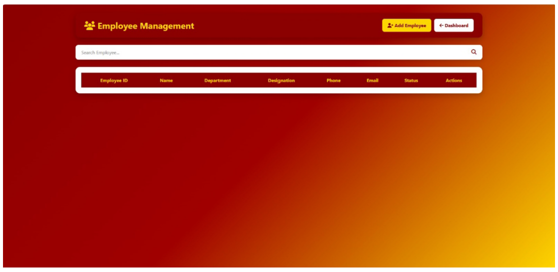
Dashboard



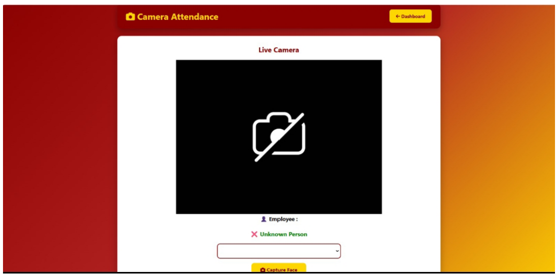
Employee Registration



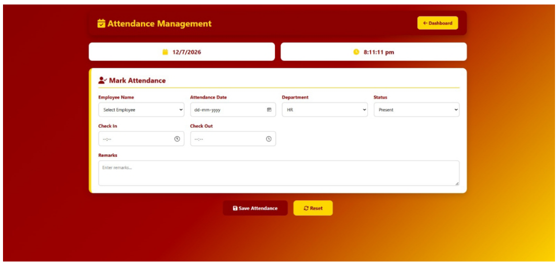
Employee management



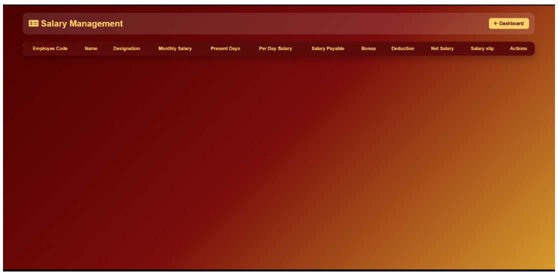
Camera Attendance



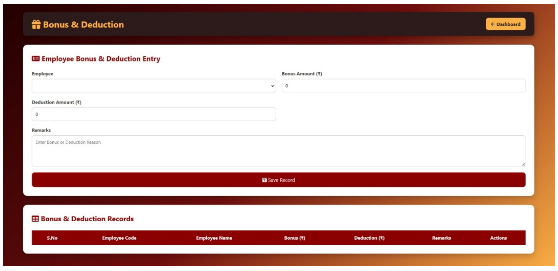
Attendance registration (manual)



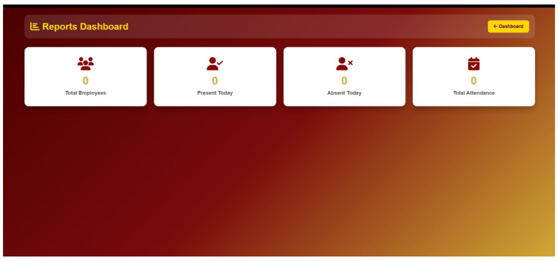
Salary Management



Bonus and deduction management

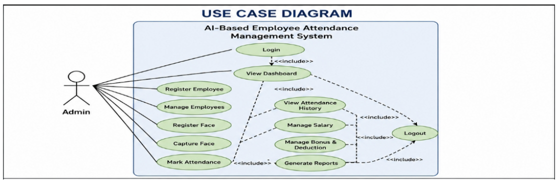


Report Generation

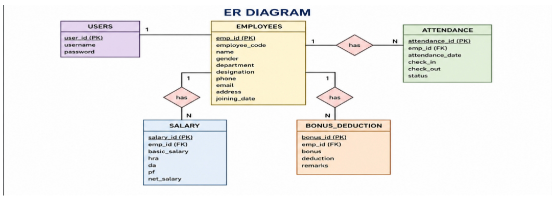


DIAGRAMS

Use case diagram



ER diagram



Relationships:

One Employee → Many Attendance Records
One Employee → One Bonus & Deduction Record

SOFTWARES USED

- Visual Studio Code
- Python 3.12
- Flask Framework
- MySQL Database
- OpenCV
- Face Recognition Library
- NumPy
- HTML5
- CSS3
- JavaScript
- ReportLab
- Git
- GitHub

TECHNOLOGY STACK

Frontend
HTML5
CSS3
JavaScript
Jinja2 Templates

Backend
Python
Flask

Database
MySQL

AI & Computer Vision
OpenCV

Face Recognition
NumPy

Development Tools
Visual Studio Code
Git
GitHub

Deployment
Flask Local Server (Current)
GitHub (Source Code Repository)

ADVANTAGES

- Fully automated attendance system.
- High accuracy using facial recognition.
- Eliminates manual attendance.
- Prevents proxy attendance.
- Easy employee management.
- Fast attendance marking.
- Generates PDF reports.
- User-friendly interface.
- Secure administrator login.

DISADVANTAGES

- Requires good lighting for accurate face recognition.
- Performance depends on camera quality.
- Internet/cloud features are not included.
- Face recognition may be affected by masks or significant facial changes.

Conclusion

The AI-Based Employee Attendance Management System successfully automates attendance using facial recognition technology. It reduces manual work, increases accuracy, prevents proxy attendance, and efficiently manages employee information, salary, and attendance reports. The project demonstrates the practical application of Artificial Intelligence in real-world attendance management.

Future Scope

- Cloud deployment
- Mobile application
- GPS attendance
- Email notifications
- SMS alerts
- Real-time analytics
- Deep Learning-based recognition
- Multi-camera support

References

- Python Documentation
- Flask Documentation
- OpenCV Documentation
- Face Recognition Library Documentation
- MySQL Documentation
- ReportLab Documentation

PROJECT / GITHUB LINK

https://github.com/priyasuja2008/AI-BASED-EMPLOYEE-ATTENDANCE-AND-SALARY-MANAGEMENT-SYSTEM-PROJECT

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